

Heat-Map Intervention Model — backend formulas

Status: implemented · costs calibrated 2026-07-24 · physics validated against satellite measurement 2026-07-27 (§0.1). The forcing block (§4) was rewritten after validation: computed sky temperature, night ET gate, day/night anthropogenic heat split, and real solar geometry for the calibration. Constants were NOT refitted — the least-squares fit failed its acceptance bar with every parameter pinned to a bound, so the model keeps its literature-derived values and the fit output carries an explicit do-not-ship flag. See [green-score-methodology.md](#) §5A. Companion: [heat-map-feature.md](#) (architecture) · [src/scripts/climate-engine/types.ts](#) (ABI + stability math, self-checked). Sources: every formula cites the methods deepsearch (2026-07-24); URL list at the end.

0 · Model class & what it may claim

This is a **2D energy-balance screening model** (same family as SUEWS/UMEP-class planning-support tools, not ENVI-met/CFD [7]). To a municipality it may honestly claim: **relative** scenario comparison, hotspot location, spatial prioritisation, order-of-magnitude ΔT . It may **not** claim absolute air temperature, hourly forecasts, thermal comfort (Tmrt/UTCI), 3D shadowing, or any single "we will cool your ward by X.X °C". Output labels: layers from footprints/OSM = *measured inputs*; ΔT fields = *modelled relative estimate (2D energy-balance screening)*; kernels = *literature-based*.

0.1 Measured accuracy — the claim boundary is now a number

Validated against 49 usable NASA ECOSTRESS scenes over Kolkata (2024-01 → 2026-07). Full method in [green-score-methodology.md](#) §5A; the figures the interface displays live in [climate-engine/accuracy.ts](#).

Phase	scenes	best achievable on this forcing	this model	shown in UI
Night	20	2.18 °C	3.13 °C	± 3.5 °C, labelled <i>calibrated</i>
Day	12	3.33 °C	4.61 °C	± 5.0 °C, labelled <i>indicative only</i>

"Best achievable" is the leave-one-out error of an unconstrained empirical predictor on the same weather inputs — an upper bound on **any** model driven by this forcing. Night leaves real headroom. **Daytime does not**: noon surface temperature turns on cloud timing, local sunshine and soil moisture that 50 km reanalysis cannot resolve, so no physics change moves it. The only lever is higher-resolution forcing (ERA5-Land at 9 km). Consequently the model **may not** be presented as quantitative in its daytime phase. The interface enforces this rather than relying on the reader.

1 · Core field equation (implemented — `types.ts`, `sim-gpu.ts`)

Per-cell temperature `T` on an $N \times N$ grid ($N=192$, ward 1400 m $\rightarrow$ **`dx = 7.29 m/cell`**), with input layers `albedo`, `veg`, `built`, `water` $\in [0,1]$:

$$dT/dt = D \nabla^2 T + S \cdot (1 - \text{albedo}) \cdot \text{sun} - k_{\text{Rad}} \cdot (T - T_{\text{sky}}) - L \cdot \text{veg} - h \cdot \text{wind} \cdot (T - T_{\text{air}}) + Q \cdot \text{built} \quad (1)$$

Explicit Euler; `dt` auto-clamped by `stableDt()` to $\min(0.25/D, 1.8/k)$ where the total relaxation rate is

$$k = k_{\text{Rad}} + h \cdot \text{wind} \quad (2)$$

Water cells relax toward `Tair - 1.5` (thermal-mass shortcut, `sim-gpu.ts` frag).

2 · Cooling spread — the empirical influence kernel λ (the key design rule)

Reframed 2026-07-25. `D` is not a thermal diffusivity and this document previously implied it was. Lateral heat conduction between 7.29 m cells is physically negligible — the diurnal damping depth in soil is ≈ 0.12 m, some eight orders of magnitude short. Real horizontal coupling in the urban surface layer comes from **advection**, which is directional and wind-dependent, not isotropic Fickian diffusion. `D` is an **empirical spatial-influence kernel**; λ , not `D`, is the meaningful quantity. λ is kept deliberately short (≈ 47 m) because this field is land-surface temperature, which real thermal imagery shows as sharp; the 120–300 m park-cooling distances in the literature are an air-temperature phenomenon.

Do not add a separate distance-decay kernel. Equation (1) already produces one: its steady state is a *screened diffusion* (modified-Helmholtz) equation whose point-response decays as $K_0(d/\lambda) \approx e^{(-d/\lambda)}/\sqrt{d}$ — the **same exponential family that dominates empirical park-cool-island measurements** (68 % of cool-core parks fit exponential decay [11]; Hanoi UGS decay vanishing ~ 800 m [12]). Adding an explicit $e^{(-d/L)}$ kernel on top would **double-count** [research §A]. Instead, tune the emergent length:

$$\lambda = \sqrt{(D/k)} \quad [\text{cells}] \rightarrow \lambda_m = dx \cdot \sqrt{(D/k)} \quad (3)$$

Calibration target: empirical decay lengths $L \approx 60\text{--}150$ m, max measurable reach $2\text{--}3 \cdot \lambda$; **Kolkata-specific:** max cooling distance ≈ 420 m, TVoE ≈ 0.77 ha [4].

LST-contrast caveat (learned 2026-07-24): a *single* diffusion constant cannot both (a) spread a park's cooling ~ 90 m AND (b) preserve the sharp hot-roof/cool-street contrast that a **Land-Surface-Temperature** field must show. Buildings here are fine-grained (interspersed with streets at grid scale), so a large D averages every hot rooftop into its cool neighbours $\rightarrow$ the field homogenises to its mean and **no reds survive** (observed at $D=9$: peak stuck $< 40^\circ\text{C}$, 0 % area > 40 , a washed-out map). The 89–420 m park "reach" in the literature is largely an **air-temperature** phenomenon; our field is labelled LST, which real imagery shows as *sharp*. So we prioritise contrast.

Chosen constant: $D = 2.5 \text{ cell}^2\cdot\text{u}^{-1}$ (was 0.15; briefly 9.0). $k \in [0.032, 0.12] \rightarrow \lambda = \sqrt{(D/k)}\cdot dx \approx 33\text{--}65 \text{ m}$ — a park still gets a near-field cooling halo (within Mitra's 67–81 m *mean* cooling distance [5]) while the dense core keeps its red hotspots. Verified: at live ambient $\approx 30^\circ\text{C}$, base mean $\approx 39^\circ\text{C}$, **$\sim 49\%$ of the core $> 40^\circ\text{C}$** , UHI $+14^\circ$ — a legible heat map. We do **not** claim the full 420 m LST reach (that needs advection we omit).

Quantity	Value	Note
CFL dt	$0.25/2.5 = 0.10$	<code>stableDt</code> clamps automatically
convergence burst	<code>sim.step(1, RESET_BURST=600)</code> per reset	$\sim$ few ms on the offscreen GPU; live stepping polishes
baseline <code>eqMean</code> validity	unchanged	boundaries $\approx$ no-flux $\Rightarrow$ diffusion approx. preserves the domain mean

3 · Per-slider spatial recipes

3.1 Tree canopy corridors (0–50) $\rightarrow$ `veg` along real OSM roads

Street trees: mature crown $\approx 100 \text{ m}^2$ ($r \approx 5.6 \text{ m}$), municipal spacing $s \approx 8\text{--}12 \text{ m}$ [8] $\rightarrow$ linear canopy $100/s \approx 8\text{--}12 \text{ m}^2$ per metre of street.

```
corridor cells = road centerline buffered  $\pm 1$  cell (buffer width  $w \approx 2\text{--}3$  cells  $\approx 15\text{--}22 \text{ m}$ )
 $\Delta\text{veg\_cell} = \min(\text{CAP}, 100/(s\cdot w\_m)) \approx 0.45\text{--}0.70$  with  $\text{CAP} = 0.7$ 
(4)
slider  $n \in [0,50]$  activates  $n/50$  of total corridor-km, HOTTEST-FIRST
```

Hottest-first: rank road segments by base-field temperature at segment midpoint — targeted greening of 1.5 % of a street network cut heat-exposed distance 19 % vs uniform [9]; ranking is precomputed per ward, deterministic. Saturation is *inherent* (veg CAP + finite road-km), no extra curve needed. **Data:** OSM highway $\in \{\text{motorway...residential, living_street, pedestrian, service}\}$ centerlines per ward — **fetched** $\rightarrow$ `previews/heat-map-3d/data/{ward}-roads.json` (500/279/532 ways, class-weighted width; ODbL attribution on-page).

3.2 Cool-roof albedo (0–100 %) → albedo on built cells only

LBNL values [6]: dark roof $\alpha \approx 0.10$ – 0.20 ; fresh white 0.70 – 0.85 ; **aged 0.55 – 0.65** . Use the honest **aged** value:

```
 $\Delta\text{albedo\_cell} = \text{built\_cell} \cdot (\text{slider}/100) \cdot (\alpha_{\text{cool}} - \alpha_{\text{base}}), \quad \alpha_{\text{base}} = 0.15, \alpha_{\text{cool}} = 0.60$  (5)
```

Maps 1:1 onto the real Microsoft-footprint raster (rasterBase built mask). The slider is "fraction of roof stock treated" — linear by definition; diminishing *returns in °C* emerge from (1) (albedo term only acts where sun hits built cells).

3.3 Pocket parks / wetlands (0–10) → veg / water patches on real open land

Kolkata evidence [4]: cool-island intensity $\text{UCI} = a \cdot \ln(\text{Area}) + b$, efficiency threshold **TVoE ≈ 0.77 ha**, max intensity 4.83 – 8.07 °C, reach ≤ 420 m. Several medium patches beat one big one [5].

```
blob radius r = 50 m ( $\approx 0.77$  ha – Kolkata's TVoE = the efficient park size)
inside blob: park:    veg = max(veg, 0.90), albedo = max(albedo, 0.20)
               wetland: water = 1, albedo = 0.06
(6)
placement: rank coarse cells by open-land fraction (1 – built density), enforce  $\geq 180$  m
           separation ( $\approx$  mean spillover distance [5]); deterministic per ward
```

Consistency check (make it an assert): local equilibrium drop inside a park $= L \cdot 0.9/k = 0.5 \cdot 0.9/0.06 = 7.5$ °C — inside Mitra's measured 4.83 – 8.07 °C band [4] with default params. The model reproduces the Kolkata literature *before* calibration. Spread beyond the blob comes from λ (§2), max reach $2-3\lambda \approx 190$ – 370 m $\approx$ the 420 m cap [4].

3.4 Vertical green facades (0–15) → effects on built cells (facades $\neq$ ground veg)

Facades cool by wall shading + ET (wall-surface -13 – 20 °C locally; ward-scale $\ll 1$ °C) [15]. Honest mapping — **never add ground veg**:

```
f = slider/15    (fraction of stock retrofitted)
Q_eff,cell = Q·built·(1 –  $\beta_q \cdot f$ )     $\beta_q = 0.03$  ← CITED (was 0.30, uncited) (7)
```

Corrected 2026-07-25. β_q was 0.30 with no source. Gunawardena & Steemers 2023 (*Buildings & Cities*, 10.5334/bc.282) is the only neighbourhood-scale measurement found: green facades cut space-conditioning energy 2.1 %, living walls 5.2 %, and moved heat-island intensity 1.86 K $\rightarrow 1.81$ K (~ 3 %). The dramatic -13 to -20 °C figures are *local wall surface* effects; the same authors found the vapour flux "advects away to background levels". **The $\Delta\text{veg_built}$ term is deleted** — its $\eta = 0.15$ had no measurement support, and adding ground vegetation for a wall treatment was double-counting. Facades act on Q only, and are now

correctly shown as a marginal ward-scale lever. $\beta q/\eta$ are ET/shading coefficients, not costs — no cost source covers them; set so ward-scale facade effect stays $\ll 1^\circ\text{C}$, consistent with the wall $-13\text{--}20^\circ\text{C}$ local-only evidence [15]. Facade ₹ cost is now solid* (₹9,500/m², §5). Revisit $\beta q/\eta$ only if a facade micro-study is later commissioned.

$P \cdot h_g/A_{fp}$ (perimeter $\times$ greened height / footprint area) is computable per ward from the real footprints + heights already in `data/*.json`.

4 · Scenario forcing (implemented; constants flagged)

```

peak 13:00:  sun = 1·(1 - 0.6·cloud_live)    tAir = tAir_live + Δpath
              tSky = skyTemperatureC(tAir, rh, cloud)
              L    = L0·evap(rh)              Q = Q0
night 22:00:  sun = 0                        tAir = tAir_live - 2.5 + Δpath
              tSky = skyTemperatureC(tAir, rh, cloud)
              L    = L0·evap(rh)·0.10·gate    Q = Q0·0.50
Δpath: 2025 = 0 · SSP2-4.5 '50 = +1.25 · SSP5-8.5 '80 = +4.1 ← CITED (Dhara et al.
2025,
          PLOS Climate 4(11):e0000724, post-AR6 India update). The former -1.2 °C "target"
          pathway is DELETED: no emissions scenario produces regional cooling over India, so
a
          negative delta on a physical-temperature axis reads as a forecast and cannot be
one.
wind_sim = clamp(wind_live/3, 0.3, 2.5)      (3 m/s ≈ sim wind 1)

```

4.1 Sky temperature is computed, not fixed (sky.ts)

`tSky` was hard-coded at **17 °C day / 11 °C night**. Those are DRY-sky values, used unsourced in a humid tropical city where the thermal-IR window is close to opaque and the effective sky sits only a few K below air. They are now replaced by **Brutsaert (1975)** clear-sky emissivity with a cloud screen:

```

e    = 6.112·exp(17.67·T/(T+243.5)) · rh/100                (hPa, Magnus)
ε    = min(1, c·(e/T_K)^(1/7) + 0.9·(1 - c·(e/T_K)^(1/7))·cloud)  c = 1.24
tSky = ε^(1/4)·T_K - 273.15                                   (10)

```

At Kolkata night conditions (28 °C, 80 % RH) this moves `tSky` **11 → 19.60 °C**. Cross-checked against Berdahl–Martin and Prata: all three agree within 0.6 °C.

The sky term cancels exactly in any urban–rural difference — both cells share `tSky`, so it sets the temperature *level*, not the *contrast*. That is the wiring test: after this change absolute temperatures rise and SUHII must not move. Measured delta was 3.55e-15 K (day) and 0 K (night).

4.2 Night evapotranspiration (`NIGHT_ET_FRACTION = 0.10` , dewpoint taper)

Stomata close after dark; what remains is soil and canopy-interception evaporation, which eddy-covariance studies put at **5–15 % of midday**. Running the full daytime `L` overnight gave parks a cooling advantage they cannot physically hold.

ET is additionally tapered to zero as the surface approaches the dewpoint, because evaporation is driven by vapour-pressure deficit and that closes continuously:

$$T_d = 243.5 \cdot \gamma / (17.67 - \gamma), \quad \gamma = \ln(\text{rh}/100) + 17.67 \cdot T / (T + 243.5) \quad (11)$$

$$L_{\text{night}} = L_0 \cdot \text{evap}(\text{rh}) \cdot 0.10 \cdot \text{clamp}((T_{\text{dry}} - T_d)/1.0, 0, 1) \quad (12)$$

`T_dry` is the no-ET equilibrium of a representative vegetated cell — the taper depends on the answer it modifies, so it is evaluated once per frame rather than per cell (cells span ~ 2 K at night; only those straddling the dewpoint would differ).

A hard cutoff was tried first and rejected: it put a 0.388 K step on park cells at $\text{RH} \approx 86\%$, inside the 85–95 % band Kolkata monsoon nights occupy, so the live humidity feed would have visibly jerked the map. The taper reduces the worst step to 0.024 K, and `assertInterventionLogic()` now sweeps RH 40–99 % asserting no step exceeds 0.1 K.

4.3 Anthropogenic heat day/night (`Q_NIGHT_RATIO = 0.50`)

Traffic, air-conditioning and commercial load all fall after dark but none reach zero. 0.50 is the round midpoint of published diurnal Q profiles for South and East Asian cities — an estimate with bounds, not a measurement.

4.4 Solar geometry (`solarElevationFactor` , used by the calibration only)

`sun = 1` is correct at local solar noon and nowhere else. ECOSTRESS's day/night flag means only "sun above the horizon", so calibration scenes span **07:06 to 17:24** local solar time and comparing them against a full-sun configuration would push solar-geometry error into the fitted constants:

$$\begin{aligned} \delta &= \text{Spencer (1971) declination series in day-of-year} \\ h_a &= (h_{\text{solar}} - 12) \cdot 15^\circ && (\text{local solar time} \Rightarrow \text{no equation-of-time term}) \\ \text{sun} &= \max(0, \sin \varphi \cdot \sin \delta + \cos \varphi \cdot \cos \delta \cdot \cos h_a) && (13) \end{aligned}$$

Measured against real overpasses, `sun = 1` overstated insolation by **2.9×** at 07:06 and **6.4×** at 16:42 in January.

Deliberately not wired into `currentParams` . The UI's "peak" phase means solar noon, where `sun  $\approx$  1` is right. Per-scene geometry belongs to the fit, not the product — same module, two consumers.

Live ambient: Met Norway per-ward; feels-like via NWS Rothfusz (see `heatIndexC`).

5 · Green Score (0–100) — grounded, not invented

All the established indices — Berlin **Biotope Area Factor** [1], **Seattle Green Factor** [2], Singapore **Green Plot Ratio** [3], Malmö GSF — share one shape: a **weighted-area ratio** $\sum(w_i \cdot A_i)/A_{\text{domain}}$. Our score blends that with cooling achieved and budget efficiency:

```
GreenScore = 100 · clamp( [ min(1,G/G_ref) + min(1,ΔT_cool/ΔT_ref) + E ] / 3 , 0, 1)
(8)
```

```
G = mean over domain of g_cell
g_cell = clamp( 1.0·veg_ground + 0.6·Δveg_corridor + 0.6·facadeGreen·built
               + 0.1·coolRoof·built + 0.8·water , 0, 1)
(9)
```

Weights consolidated from the verified tables (Berlin: in-ground 1.0, wall 0.5, roof 0.7, sealed 0.0 [1]; Seattle: 0.0–0.7 [2]): **sealed 0.0 · cool-roof 0.1 · lawn 0.2 · shrub 0.3 · tree canopy 0.6 · green facade 0.6 · water/wetland 0.8 · in-ground park 1.0.**

- **G_ref = 0.45** — mid of Berlin's own 0.30–0.60 target band [1]
- **ΔT_cool = eqMean(baseline) – mean(T)** (already computed) with **ΔT_ref = 2.5 °C** (ward-scale cooling beyond ~2–3 °C is not credible [research §B])
- **E = budget efficiency = clamp((ΔT_cool / cost_₹crore) / E_ref , 0, 1)**, provisional **E_ref = 0.15 °C per ₹ crore. This is TOOL-RELATIVE, not an external benchmark** — no published cost-per-degree-cooled figure exists to normalise against, so a ward can only score well relative to what this model can produce. Labelled as such rather than presented as grounded.

Weights corrected 2026-07-25: equal thirds, was 40/40/20. The old split had no published precedent and implied greening and cooling were each exactly twice as important as cost. Equal weighting is the most commonly applied approach in composite-indicator practice (OECD/JRC 2008, *Handbook on Constructing Composite Indicators*) and the convention in urban-resilience indices. An arbitrary split dressed as a derived one is worse than a plain citable one. - **UI must expose the three sub-scores raw** (greening ratio, °C cooled, °C per ₹ crore) — transparency is what makes a BAF-style score trusted [1,2].

Budget (binding) — India ₹ unit costs [C1–C6, all cited]

cost = $\sum$ (slider activation × physical_quantity × unit_cost_₹). Quantities derive from the real footprints/roads/heights already in **data/*.json**.

Slider	Physical unit	Unit cost ₹ (range)	Cost formula
Cool-roof albedo (0-100 %)	per m ² roof treated	₹150 /m ² coating installed (<i>budget lime-wash</i> ₹40 /m ² ; range 16–270) [C1,C2]	$(\text{slider}/100) \cdot \sum \text{roof_area_m}^2 \cdot \text{rate}$
Tree corridors (0-50)	per tree, ~110 trees/km	₹1,500 /tree established municipal (<i>range</i> 500–3,000) → ≈ ₹1.65 L/corridor-km [C3]	$\text{active_km} \cdot 110 \cdot \text{rate}$
Pocket parks (0-10)	per 0.785-ha blob (r 50 m)	₹1.5 cr/ha → ≈ ₹1.18 cr/blob (<i>bare park</i> ~½; range 1.1–2.25 cr/ha) [C4]	$\text{n_blobs} \cdot 0.785 \cdot \text{rate}$
Green facades (0-15)	per m ² wall greened	₹9,500 /m ² (<i>range</i> 2,700–17,000) [C5]	$\text{greened_facade_m}^2 \cdot \text{rate}$

Cost-effectiveness realism (bake this in): cool roofs (~₹150/m²) are ~60× cheaper per m² than green walls (~₹9,500/m²), parks in between (~₹110–225/m² of park). So the ₹-per-°C knob *naturally* makes cool roofs the dominant lever — which **matches real Indian Heat Action Plan practice** (Ahmedabad/Telangana lead with cool roofs) [C1,C6]. A defensible emergent behaviour, not a designed-in bias. **Wetland ₹/ha** — no open source found; uses the same park ₹/ha as a **flagged placeholder**.

Cited display anchors (UI "did you know" / tooltip copy — every one sourced)

Claim	Figure	Source
Cool-roof surface-temp drop	up to –30 °C (lime white-wash)	Ahmedabad HAP / NRDC [C2]
Cool-roof indoor air drop	–2.1 to –4.3 °C vs traditional roof	NRDC / Telangana policy [C1,C2]
Street trees, afternoon air	–5.6 °C (tree-lined vs treeless road, Bangalore); road surface –27.5 °C	WRI India [C3b]
Canopy dose-response	each +10 % tree cover ≈ –0.3 °C ambient	WRI India [C3b]
Ahmedabad HAP outcome	>1,100 heat deaths avoided/yr; –27 % mortality on ≥45 °C days	PreventionWeb / NRDC [C6]

These are the "pitch slide" numbers — display them near the relevant slider, cited, so the tool teaches while it runs. They are *literature anchors*, not this model's outputs (keep that labelling per §0).

Diminishing returns (display form)

For marginal-cooling readouts (not the physics — the physics saturates on its own): $\Delta T(n) = \Delta T_{\text{max}} \cdot (1 - e^{(-k_s \cdot n)})$ — the canonical saturating form; canopy cooling accelerates only above ~40 % cover and saturates near 0.8 [18].

6 · Derived readouts (implemented — definitions of record)

```

UHI Δ      = mean(T) - T_rural_ref
T_rural_ref = (S·(1-0.25)·sun - L·1 + kRad·Tsky + h·wind·Tair) / k      (veg=1, built=0
cell
           under the SAME live forcing – never a stale constant)
Area > 40 °C = fracAbove(40) from sim.stats()
Histogram    = 12 bins over T ∈ [26, 48] °C

```

7 · Symbol ↔ code map

Formula	Code	File
(1)(2) dt clamps	<code>SimParams</code> , <code>stableDt</code> , <code>cflDt</code> , <code>decayDt</code>	<code>climate-engine/types.ts</code>
(1) GPU step	<code>STEP_FRAG</code>	<code>climate-engine/sim-gpu.ts</code>
λ retune	<code>currentParams</code> D: 0.15 → 2.5 + reset burst	<code>types.ts</code> + instrument <code>resetSim()</code>
(4) corridors	<code>applyInterventions</code> + new <code>roadMask</code>	instrument (pending build)
(5) cool roofs	<code>applyInterventions</code> albedo term (retune 0.0032 → eq. 5)	instrument
(6) parks	<code>applyInterventions</code> park loop (r 63 m → 50 m; placement → open-land rank)	instrument
(7) facades	<code>applyInterventions</code> + <code>currentParams</code> Q_eff	instrument
(8)(9) score	replaces <code>score = round(cooling·38)</code>	instrument <code>refreshStats()</code>
§4 forcing	<code>currentParams()</code> , <code>currentParamsForReference()</code>	<code>climate-engine/heat-map-model.ts</code>
(10) sky temperature	<code>skyTemperatureC</code> , <code>saturationVapourPressure</code>	<code>climate-engine/sky.ts</code>
(11)(12) night ET taper	<code>dewpointC</code> , <code>nightLatent</code> , <code>NIGHT_ET_FRACTION</code> , <code>DEWPOINT_TAPER_K</code>	<code>sky.ts</code> + <code>heat-map-model.ts</code>
§4.3 night Q	<code>Q_NIGHT_RATIO</code>	<code>heat-map-model.ts</code>
(13) solar geometry	<code>solarElevationFactor</code>	<code>climate-engine/sky.ts</code> (calibration only)
§0.1 accuracy bands	<code>ACCURACY</code> , <code>bandLabel</code>	<code>climate-engine/accuracy.ts</code>
§6 UHI	<code>refreshStats()</code> rural ref	instrument

New asserts when implemented: $\lambda(D=2.5, k=0.06) \approx 47$ m; park local drop $\in [4.8, 8.1]$ °C (Mitra band [4]); eqMean vs converged diffused mean drift < 0.3 °C; corridor $\Delta\text{veg} \leq \text{CAP}$

Self-checks now in place (all runnable with `node --experimental-strip-types`):

Check	Asserts	Where
<code>assertSkyLogic()</code>	<code>tSky < tAir</code> always; rises with humidity and cloud; 19–21 °C at 28 °C/80 %; dewpoint ≈ 24.2 °C at 28/80; sun down at 01:00 and 23:00; ≈ 1 at midsummer noon; ≈ 0.24 at equinox 07:00	<code>sky.ts</code>
<code>assertInterventionLogic()</code>	cool roofs and trees cool the mean; built-core equilibrium 40–50 °C; cost monotonic; score bounded; night ET varies smoothly with humidity (no step > 0.1 K over RH 40–99 %); ET reaches exactly zero below the dewpoint	<code>heat-map-model.ts</code>
<code>assertAccuracyLogic()</code>	model error never beats the data ceiling; displayed band never understates measured error; daytime never presented as more certain than night	<code>accuracy.ts</code>
<code>validate-model.mjs</code>	8 checks against published benchmarks, 8/8 passing	<code>scripts/</code>

8 · Display pipeline (how the field becomes colour — implemented)

One texture, two channels, three consumers:

```
bridgeField(): R = 3×3 box-blur(T) → ground drape (smooth gradients, no cell dither)
              G = raw T           → building tint (true per-building temperature)
readouts/stats: raw sim field directly (never the blurred copy)
```

- **Buildings sample ONCE per building** at the footprint centroid (`aCtr` `vec2` attribute, sampled in the vertex shader → flat varying). One building = one thermal object = one colour; kills per-pixel hot/cool checkering across a facade.
- **Tint modes** (`uTintMode` , UI chip "Gradient | 5-Class"): 1 = continuous ramp (default); 2 = snapped to the 5 legend classes (`t` → `.17/.48/.70/.85/.97`) — choropleth read, buildings visibly change class as interventions cool them. Mode 0 (per-pixel) retired.
- Roof-speckle detail is damped $0.35\times$ in per-building modes (it compounded the noise).

9 · Citations (from the methods deepsearch, fetched 2026-07-24)

1. Berlin Biotope Area Factor — formula + weights. ugl.sg/wp-content/uploads/2021/01/20191002_biotope_area_factor.pdf
2. Seattle Green Factor score sheet. app.dcoz.dc.gov/Exhibits/2010/ZC/08-06-9/Exhibit14.pdf
3. URA Singapore — Green Plot Ratio / LUSH. ura.gov.sg/guidelines/development-control/.../greenery/

4. Mitra et al. 2022 (Frontiers Env. Sci.) — tropical megacities incl. **Kolkata**: $UCI = a \cdot \ln A + b$, TVoE 0.77 ha, reach 420 m, 4.83–8.07 °C. frontiersin.org/articles/10.3389/fenvs.2022.1073914/full
5. Zhengzhou park spillover 2023 (Frontiers Earth Sci.) — cooling distance mean 179 m, ~ 1 °C/100 m, patch-splitting result. frontiersin.org/articles/10.3389/feart.2023.1133901/full
6. LBNL Heat Island Group — roof albedo values. heatland.lbl.gov/coolscience/cool-roofs
7. UMEP docs (SOLWEIG/SUEWS) — model-class positioning. umep-docs.readthedocs.io
8. Arboriculture & Urban Forestry 2021 — street-tree spacing/crowns. auf.isa-arbor.com/content/47/5/183
9. arXiv 2512.11753 (2025) — targeted street greening beats uniform (1.5 % $\rightarrow$ –19 %). arxiv.org/abs/2512.11753
10. Blue-green corridors (PMC8622358) — corridor cooling to 600–750 m, optimal width 20–35 m.
11. Yang et al. 2026 (SCS) — 68 % of parks: exponential decay (abstract-verified). sciencedirect.com/science/article/abs/pii/S2210670726002714
12. Nguyen et al. 2025 — Hanoi exponential decline (abstract-verified). sciencedirect.com/science/article/pii/S2590252025000789
13. Vertical greening (Elsevier set) — wall –13–20 °C local (abstract-verified). sciencedirect.com/science/article/pii/S0378778825015142
14. Nature Communications 2021 — 40 % canopy threshold/saturation. nature.com/articles/s41467-021-26768-w

Cost + India-anchor sources (calibration agent, 2026-07-24)

- **C1.** Telangana Cool Roof Policy 2023-2028 — lime wash $\sim$ ₹1.5/sq ft; indoor –2.1–4.3 °C. telangana.gov.in/wp-content/uploads/2023/05/Telangana-Cool-Roof-Policy-2023-2028.pdf
- **C2.** NRDC "Keeping It Cool: How India Can Protect... with Cool Roofs" — Ahmedabad white-lime ₹0.50/sq ft; surface up to –30 °C. nrdc.org/sites/default/files/keeping-it-cool-roofs-india-fs.pdf
- **C3.** Grow Billion Trees 2025 — itemised India tree plant+maintain cost build-up (₹120–800 plant; ₹1,000–3,000 3-yr maintained; Miyawaki ₹500–1,200/m²). growbilliontrees.com/pages/what-is-the-cost-of-planting-and-maintaining-a-tree-in-india
- **C3b.** WRI India, "Urban Trees' Cooling Potential" — Bangalore –5.6 °C air / –27.5 °C surface; +10 % canopy $\approx$ –0.3 °C. wri.org/insights/urban-trees-cooling-potential
- **C4.** Gujarat AMRUT 2.0 urban gardens (ANI, Feb 2026) — Bhavani Garden ₹1.26 cr/1.09 ha; Kailash Vatika ₹2.25 cr/ha. aninews.in (Gujarat 131 urban gardens)
- **C5.** IndiaMart / Bricknbolt vendor ranges — green wall ₹850–975/sq ft typical (₹2,700–17,000/m²). indiamart.com green-wall listings
- **C6.** PreventionWeb / NRDC — Ahmedabad HAP: >1,100 deaths avoided/yr, –27 % mortality on hottest days. preventionweb.net (India pioneering Heat Action Plan)

Confidence (agent's own flags): cool-roof ₹/m² = HIGH (state policy + NRDC + vendor converge); tree component costs = MED-HIGH (municipal spec > the ₹299 subsidised NGO package); park ₹/ha = MED (2 named Gujarat gardens, fully-appointed — bare park cheaper); green-wall ₹/m² = MED (commercial vendor, excludes irrigation opex); wetland = placeholder (no source).

(Methods citations 1–18 above match the methods-agent report; abstract-only sources flagged — re-verify before quoting verbatim in a client deliverable.)